

Testing Circuit & PCB

5 December 2025

- The circuit ~~and~~ as designed on the current PCB has not been tested. While the board is designed for a Particle Borne, we can use jumper wires and a 1310 to verify that the board works without needing to worry about particle.

Connections:

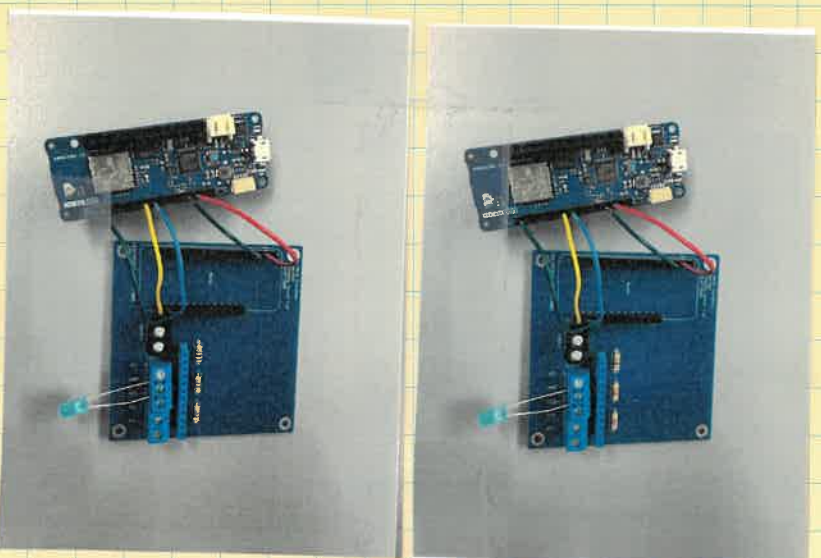
Arduino 3V3 to Borne Left Header 2
 Arduino GND to Borne Left Header 4
 Arduino 5V to Borne Right Header 11
 Arduino SDA to Borne Right Header 12
 Arduino DG to Borne Right Header 9

- Initial test of main PCB was successful. Used a basic Blink sketch on D10 to test LED

```

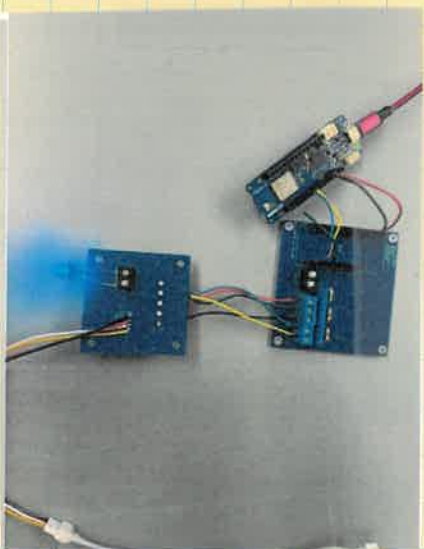
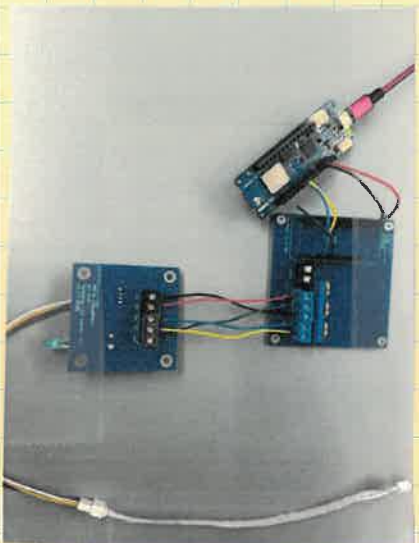
void setup ()
{
  Serial.begin (9600);
  pinMode (6, OUTPUT);
}

void loop ()
{
  digitalWrite (6, HIGH);
  delay (500);
  digitalWrite (6, LOW);
  delay (500);
}
  
```

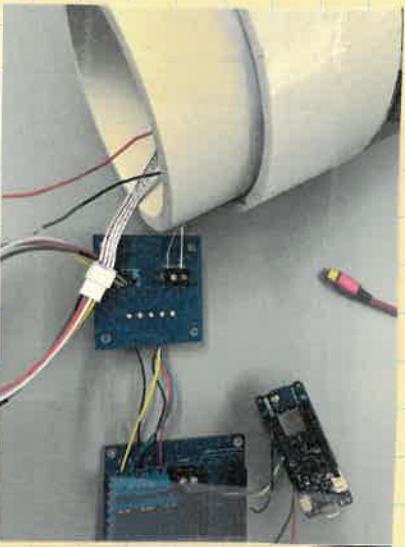


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- After verifying the main board works as expected, moved to add in the secondary board. Repeated the same connection as before, and connected the main board and daughter board. Installed LED on daughter board directly. Used same code to verify LED operation. Board worked as expected.



- Now that function of main board & daughter board is verified, connected JST from daughterboard board to IIR receiver in turbidity device. Ran existing IIC ~~Scann~~ Scanner sketch to verify connection. Scanner returned devices at 0x29, 0x60, and 0x68. When JST was disconnected it returned only 0x60 and 0x68. Confirmed with Hagar that 0x29 is the expected IIC address for the sensor.



```

--- Scan started ---
I2C device found at address 0x29 !
I2C device found at address 0x60 !
I2C device found at address 0x68 !
--- Scan finished ---

--- Scan started ---
I2C device found at address 0x29 !
I2C device found at address 0x60 !
I2C device found at address 0x68 !
--- Scan finished ---

--- Scan started ---
I2C device found at address 0x29 !
I2C device found at address 0x60 !
I2C device found at address 0x68 !
--- Scan finished ---

```

- Ready to proceed with next steps of testing. Next up is :
 - Replace MxR with Bora
 - Convert existing code to work with Particle
 - Find existing code
 - Verify JST setup is still physically able to deploy
 - learn how to use the calibration Turbidimeter
- So far it looks like we're on track to deploy a test by end of month; goal is still end of December.